Data Communication

Topic: OSI Model Simulator

B.Tech Semester IV

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OSI MODEL SIMULATOR REPORT

A Detailed Study and Simulation of Data Encapsulation and Decapsulation in the OSI Model

1. Abstract

In modern computer networking, structured communication between systems is essential for reliable data exchange. The OSI (Open Systems Interconnection) model provides a standardized framework that defines how data is transmitted across networks through seven distinct layers.

However, despite its importance, students often struggle to visualize the actual flow of data through these layers. This project presents an **OSI Model Simulator**, developed to provide a comprehensive and interactive understanding of the processes of **data encapsulation and decapsulation**.

The simulator is implemented using web technologies such as HTML, CSS, and JavaScript, and demonstrates how data is transformed at each layer during transmission. It acts as an educational tool that bridges the gap between theoretical knowledge and practical visualization.

2. Introduction

Data communication is the backbone of modern digital systems. Whenever a user sends a message, accesses a website, or streams content, data is transmitted across networks following a structured process.

To standardize this process, the **OSI model** was introduced by the International Organization for Standardization (ISO). It divides communication into seven layers, each responsible for a specific function.

Traditional learning methods focus mainly on memorization of layer names and functions. However, deeper understanding requires visualization of:

- Data transformation across layers

- Header and trailer addition
- Addressing mechanisms
- Protocol interactions

The OSI Model Simulator addresses these challenges by providing a **step-by-step graphical representation** of how data travels from sender to receiver.

3. Objectives

The main objectives of this project are:

- To provide an in-depth understanding of the OSI model
- To demonstrate encapsulation and decapsulation processes
- To simulate real-world networking parameters
- To develop an interactive educational tool
- To enhance conceptual clarity through visualization

4. Theoretical Background

4.1 OSI Model Overview

The OSI model consists of seven layers:

Name	Function
Application	Interface for user applications
Presentation	Data encoding, encryption, compression
Session	Session establishment and management
Transport	Reliable data delivery, segmentation

Network	Logical addressing and routing
Data Link	Physical addressing and error detection
Physical	Transmission of raw bits

Each layer communicates with its corresponding layer on another system using protocols.

4.2 Data Encapsulation (Sender Side)

Encapsulation is the process of adding control information to data as it moves down the OSI layers.

Steps involved:

- Application Layer: Generates raw data
- Presentation Layer: Encodes data
- Session Layer: Adds session details
- Transport Layer: Adds port numbers → Segment
- Network Layer: Adds IP address → Packet
- Data Link Layer: Adds MAC address + FCS → Frame
- Physical Layer: Converts frame into bits

Final Transformation:

Data → Segment → Packet → Frame → Bits

4.3 Data Decapsulation (Receiver Side)

Decapsulation is the reverse process:

- Physical Layer receives bits
- Data Link Layer removes frame header
- Network Layer removes IP header
- Transport Layer extracts segment
- Session and Presentation layers process data

- Application layer delivers final message

4.4 Protocols Used in OSI Layers

- Application: HTTP, FTP, SMTP
- Transport: TCP, UDP
- Network: IP
- Data Link: Ethernet
- Physical: Transmission media standards

4.5 Importance of OSI Model

- Standardizes communication
- Enables interoperability
- Simplifies troubleshooting
- Supports modular design

5. System Architecture

The simulator follows a **three-phase architecture**:

1. Sender Side (Encapsulation)
2. Transmission Medium
3. Receiver Side (Decapsulation)

The system simulates **14 steps** representing each OSI layer's role.

6. System Features

- Predefined network devices with IP and MAC
- User-defined message input

- Protocol selection (TCP/UDP)
- Encoding methods (UTF-8, ASCII, Base64)
- Step-by-step OSI layer simulation
- Automatic generation of:
    - Port numbers
    - Session IDs
    - MAC/IP addresses
    - Frame Check Sequence (FCS)
- Binary conversion at physical layer
- Visual animation of data transmission
- Reset and replay functionality

7. Working Principle

7.1 Input Phase

User provides:

- Sender device
- Receiver device
- Message
- Protocol and encoding

7.2 Sender Side Processing

Data flows from Layer 7 to Layer 1:

- Data preparation
- Encoding
- Session establishment
- Segmentation
- Routing information
- Framing
- Bit conversion

7.3 Transmission Phase

Data is transmitted as binary across the network medium.

7.4 Receiver Side Processing

Reverse process:

- Bits → Frame → Packet → Segment → Data
- Headers removed
- Data decoded
- Final message delivered

8. Implementation Details

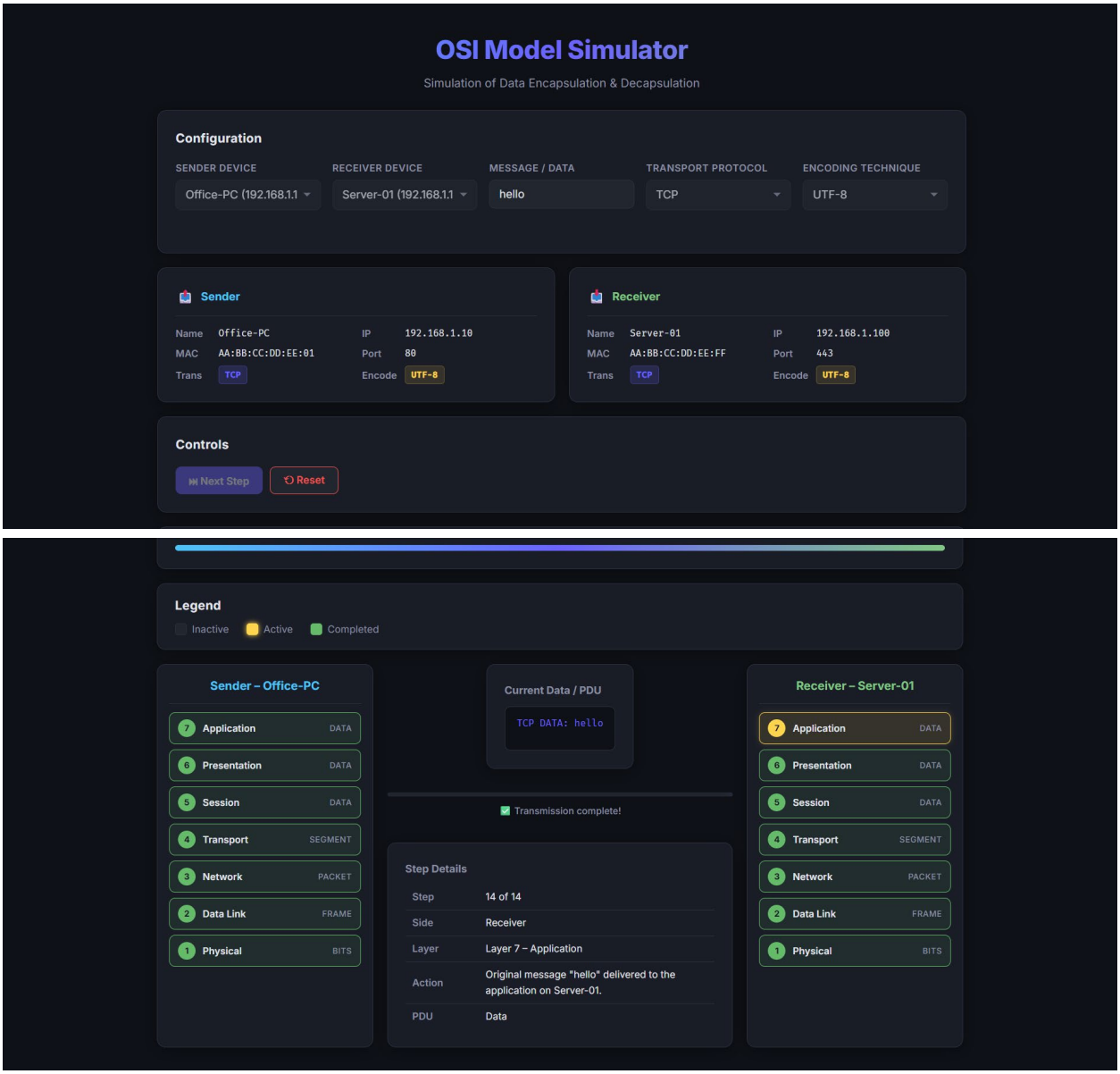
Technologies used:

- HTML – Structure
- CSS – Styling
- JavaScript – Logic and interactivity

Core implementation includes:

- Event-driven simulation
- Dynamic data transformation
- Layer-wise execution

9. Screenshots of the System



10. Project Website Link

https://github.com/jiyaa-patel/osi-encapsulation-decapsulation

Project URL:
https://osi-encapsulation-decapsulation.vercel.app/

11. Advantages

- Improves conceptual understanding
- Interactive and engaging
- Easy to use
- Lightweight and accessible
- Ideal for academic learning

12. Limitations

- Not a real network simulation
- Limited protocol coverage
- No real-time packet loss handling
- Basic visualization

13. Future Scope

- Real-time packet simulation
- Error handling (packet loss, delay)
- Advanced UI/UX improvements
- Multi-device communication
- Integration with backend networking tools

14. Conclusion

The OSI Model Simulator effectively transforms a complex theoretical concept into an interactive learning experience. By visually demonstrating encapsulation and decapsulation, it enhances understanding of network communication processes.

This project serves as a valuable educational tool for students and provides a strong foundation for advanced networking studies.

15. References

- Data Communications and Networking – Behrouz A. Forouzan
- Computer Networks – Andrew S. Tanenbaum
- ISO OSI Model Documentation
- Online educational resources